MYSX-256/512: A 256-bit Block Cipher with 512-bit Keys (v0.1-draft2)

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Abstract

We present MYSX-256/512, a 256-bit block cipher with 512-bit keys and 128 rounds. The design combines Feistel cells with Simon's round function, a Myszkowski byte transposition, word rotations, and an ARX key schedule. We provide the full specification, test vectors, and a self-analysis covering differential, linear, integral, and related-key cryptanalysis. The differential analysis reports an observed upper bound on the best single-trail weight (114 at 14 rounds, extrapolating to the 128-bit threshold at r≈15; the runs do not vary the rotation offset, so this extrapolation is likely optimistic, see §3.1). The r=4 trail has been directly verified at w=32 with a clustering factor of 2.00× (observed 2^{-7} vs MILP 2^{-8}). The linear analysis is built on an exact model of F: the LAT admits a closed form (`|corr(b,g)| = 2^{-h(b)}`, `h(b) = rank(A_b)/2` on a 32-cycle), mask propagation is bit-exact, and every trail carries an exact key-dependent sign. Within this model we verify `|corr|=1` chains over up to 9 consecutive rounds (three independent keys), and exact linear hulls at r=10,11,12 match the real cipher within sampling error ($|z| < 0.4$). A beam search over the exact model, run in both directions (forward and backward) for all 31 rotation offsets and up to 128 rounds, and merged (§4.10), gives achievable single-trail weights of 2 at 12 rounds, 5 at 14 rounds, 38 at 20 rounds, 786 at 64 rounds and 2248 at 128 rounds; after round ~20 the best weight grows by roughly 12–21 per round up to R=112 (the last step, 112→128, is ≈36 per round because only one offset fits a full 128-round trail), and the largest round count with an achievable single trail of weight < 64 (≈2^{128} texts) is 22 (weight 63, a tie with the threshold). Among the tabulated round counts, backward trails are the cheapest at R=9–14, 18, 20, 24–48 and 96–128, and forward trails at R=16, 22, 56, 64 and 80. A hull check against the real cipher shows that the offset-20 backward trail at R=14 (weight 5, covering rounds 100–113) is isolated in its hull and its empirical correlation matches 2^{-5}: this is a verified linear distinguisher on a 14-round window of the cipher (five cheap rounds followed by a 9-round weight-0 chain), not an attack on the full cipher. Other trails can be equal to their single-trail value, a factor ~2 larger, or exactly cancelled (§4.10.3). No lower bound over 128 rounds is established; the weight-0 windows are a property of 5–9 round windows, not an attack on the full cipher. A preliminary integral analysis with standard division-property rules finds integral distinguishers covering 6 rounds with a 2^{32}-text integral (word 0 active) and 5 rounds with a 2^{64}-text integral (words 0 and 4 active); a 10-round single-bit dependency propagation is also reported but is not an integral distinguisher (2 texts). Only r0=0 was tested. The key schedule was audited with 22 structural and empirical tests up to weight-6 with no persistent Δrk=0 pattern found. MYSX has not been externally audited. It is released as v0.1-draft2 for public review and must not be used to protect real data.

1. Introduction

MYSX-256/512 is an experimental block cipher. It is offered as a public proposal for cryptanalysis, not as a production primitive. The author makes no claim of security beyond the observed results reported here, and explicitly welcomes external review.

1.1. Security objectives (and non-objectives)

Model. MYSX is analyzed in the single-key, chosen-plaintext model for confidentiality, combined with an encrypt-then-MAC AEAD wrapper for integrity (Section 2.9). The design does not claim security under related-key, chosen-ciphertext, or side-channel models; see Section 5 for related-key coverage and Section 9 for engineering notes.

Claimed security level. None formally. The paper reports observed upper bounds on single-trail weights (Section 3), a verified exact linear model up to a small number of rounds (Section 4), a full-length beam search over that model in both directions (§4.10), and preliminary integral distinguishers at 5–6 rounds for multi-bit inputs (§4.9). These are inputs to cryptanalysis, not proofs.

Reading thresholds. A differential trail of weight `w` needs about `2^w` chosen pairs; a linear trail of weight `w` has `|corr| = 2^{-w}` and needs about `2^{2w}` known texts. Therefore a 2^{128}-text budget corresponds to a differential weight threshold of ≈128 but to a *linear* weight threshold of ≈64. Differential and linear weights in this paper must not be compared directly; compare `w_diff` with `2·w_lin`.

Non-objectives. MYSX does not claim:

- post-quantum security beyond the trivial Grover bound (2^{256} quantum iterations for a *uniformly random* 512-bit key; keys produced by the HKDF-based API of Section 2.10 from a 256-bit secret give 2^{128}, see Section 6);
- resistance to related-key attacks beyond the heuristic coverage in Section 5;
- resistance to integral attacks beyond the preliminary 5–6 round distinguishers reported in §4.9, nor resistance to impossible- differential, boomerang, or algebraic attacks (none analyzed);
- resistance to linear-hull effects beyond the sampled checks of §4.6 and §4.10.3 (no hull-aware bound beyond R≈14);
- constant-time execution in JavaScript (side channels out of scope).

1.2. Design rationale

MYSX combines four building blocks:

- **Simon's F** `F(x) = (rotl(x,1) ∧ rotl(x,8)) ⊕ rotl(x,2)`. Chosen for its **studied** diffusion/linear/differential properties in Simon and its simplicity (no S-boxes, only AND/XOR/rotl). It does **not** inherit Simon's security proofs; the surrounding structure is different. See Section 7 for what transfers and what does not.
- **Feistel cells over 32-bit word pairs.** Four parallel cells per round, then a byte transposition over the full 32-byte state, then word rotations. The result is

classified as an SPN-like structure with Feistel cells, not as a “2-branch Feistel”. See Section 2.1.
- **Myszkowski transposition** (Section 2.3.1). A fixed byte permutation derived from the keyword `MYSZKOWS`. The choice is didactic; it provides byte-level diffusion but no cryptographic argument beyond that. An optimized permutation would provide strictly better diffusion per round; see Section 9.
- **ARX key schedule** (Section 2.4). A bijective Feistel 8+8 update with modular addition, XOR, and rotations. Chosen for simplicity and to avoid S-boxes; analyzed in Section 5.

The parameter choices (256-bit block, 512-bit key, 128 rounds) are not derived from a formal argument. They were chosen to:

- provide a large block (avoid birthday-bound issues of 64-bit blocks);
- provide a large key (Grover leaves 256-bit security for a random 512-bit key);
- provide a large round count relative to Simon128/128 (68 rounds) as a precaution against unknown trails.

No claim is made that 128 rounds is sufficient. The round count is a parameter, not a security level.

1.3. Comparison with related designs

Design	Block	Key	Rounds	Structure	S-boxes
Simon128/128	128	128	68	Feistel	No
Speck128/256	128	256	34	ARX	No
ChaCha20	512	256	20	ARX	No
MYSX-256/512	256	512	128	SPN-like with Feistel cells	No

MYSX’s closest sibling is Simon. The differences (block size, branch organization, transposition, word rotations, non-linear key schedule) are substantial enough that Simon cryptanalysis does not transfer directly (see Section 7).

Performance. MYSX is estimated to run at a few MB/s in JavaScript, versus hundreds of MB/s for AES-GCM via WebCrypto (AES-NI). MYSX is not competitive on performance; it is a research primitive.

1.4. Versioning and revision notes

Versioning. The reference implementation (`mysx.prod.js`) is labelled `v0.1-draft` and has not changed since. This document is the second documentation draft of that same design and is therefore labelled `v0.1-draft2`. An earlier revision of this text carried the label `v0.2-draft`; that label is withdrawn because it suggested a change of the cipher or of the code, and there is none.

Changes since the previous draft (`v0.1-draft`).

- **New direction:** the beam search (§4.10) now supports `--direction backward` (the offset is the index of the *last* round of the trail, mod 31) and a `merge` command that combines results from both directions. The table of §4.10.2 is now the minimum over forward and backward and over all offsets; a few `v0.1-draft` checkpoint values (R=12, R=48, R=112, R=128, and in general those where backward wins) are superseded.
- **New verified distinguisher (on a 14-round window):** the offset-20 backward trail at R=14 (weight 5) is isolated in its hull (`hull_count = 1` trail) and its empirical correlation matches 2^{-5} on the real cipher (§4.10.3). This is the first non-trivial linear distinguisher the analysis certifies. It covers rounds 100–113 and consists of five cheap rounds followed by a 9-round weight-0 chain.
- **Correction on R=128:** in forward mode only r0=0 fits a 128-round trail. In backward mode, offset 3 gives `end_round(3) = 127`, `realizable_R = 128`, so the combined R=128 = 2248 at offset 3 backward **is a real trail inside the cipher**, not a virtual extension. The `v0.1-draft` statement that "for r0>0 the value at R=128 is a virtual extension" is forward-specific; §4.10.2 clarifies.
- **Hull check (§4.10.3)** extended: the r0=23 forward trail continues to cancel from R=8; the r0=24 forward trail cancels from R=8; the r0=12 forward trail is exact up to R=12; the offset-20 backward trail is exact from R=1 to R=14. Hull effects are not uniform.
- **Multi-bit starts:** explored (`pairs47`, `rand:5000:4`) and do not improve on single-bit starts at any R. `pairs47` is a small structured subset; `rand:5000:4` covers 5000 masks out of 2^{256} and is not conclusive for structured multi-bit masks. The honest statement is "the multi-bit starts tested do not improve on single-bit".
- Added §4.10.1 backward-step rule and its `selftest` (every forward edge is a backward edge with the same cost, 325 edges checked).
- Abstract: numbers updated to the combined forward+backward values; added the verified distinguisher at R=14.

Corrections made in `v0.1-draft2` (errors found while reviewing the previous text; no experimental result changed):

- §3.3: the direction of the "systematic model error" argument was inverted. An observed probability *higher* than predicted means the model *over*estimates the cost (true weight 7, not 9). Fixed.
- §3.2: the r=4 trail is not a row of the table; the text now says so. §3.1/§3.2/§3.4 now state that the rotation offset was not varied in the differential runs, which makes the r≈15 extrapolation optimistic.
- §4.10.2 and the abstract: the statement on which direction is cheapest contradicted the table; rewritten from the table. The "15–25 per round" growth figure did not match the table (12.5–20.9 per round up to R=112, ≈36 for 112→128); replaced.
- §4.10.3, abstract, §4.7, §10: "distinguisher at 14 rounds" now says "on a 14-round window (rounds 100–113)" and explains that the weight is flat from R=5 to R=14.
- §4.9, §7: Simon integral figures and the Simon literature list corrected against the sources (Simon32: 14 rounds with conventional bit-based division property, 15 with three subsets; the Simon128 26-round integral is Xiang et al.; ePrint 2014/973 is Shi et al.; the 49-round Simon128/128 result is a key-recovery attack

by Chen–Wang; Biryukov–Roy–Velichkov is FSE 2014).
- §4.10.4: the hull gain needed at R=24 (w=88) to reach the 2^{128} budget is 2^{24} in correlation (w − 64), not 2^{40}.
- §1.1, §6: the Grover 2^{256} figure holds for a uniformly random 512-bit key only; HKDF-derived keys give 2^{128}. The HKDF sentence in §6 was imprecise and is fixed.
- §8: functional test count aligned with `README.md` and `SECURITY.md` (17 tests).
- §9.2: note that 128-bit random nonces make accidental reuse negligible, so `NonceGuard` mainly enforces the per-key message limit.

2. Specification

2.1. Parameters

Parameter	Value
Block size	256 bits (8 words of 32 bits)
Key size	512 bits (16 words of 32 bits)
Rounds	128
Structure	SPN-like: 4 parallel Feistel cells + byte transposition
Non-linearity	`F(x) = (rotl(x,1) ∧ rotl(x,8)) ⊕ rotl(x,2)`
Diffusion	Myszkowski transposition (keyword `MYSZKOWS`)
Rotation	`rotl(word, (r mod 31) + 1)` per word and round
Final whitening	XOR with 8 words from the key schedule

Structural note. MYSX is sometimes described as a “2-branch Feistel” by analogy with Simon. Strictly, each round applies four independent Feistel cells over 32-bit word pairs, then a byte transposition over the full 32-byte state, then word rotations. It is more accurately classified as an SPN with Feistel cells. The “Feistel” label is a convenience.

2.2. Notation

- `rotl(x, n)`: circular left rotation of `x` by `n` bits.
- `∧`: bitwise AND.
- `⊕`: bitwise XOR.
- `s[r]`: state at round `r` (`s[0]` = plaintext, `s[128]` = ciphertext).
- `s[r][i]`: word `i` (0..7) of the state at round `r`.
- `rk[r][i]`: subkey `i` (0..7) of round `r`.
- `rkFinal[i]`: word `i` (0..7) of the final whitening.

2.3. Constants

```
CONST[0..7] = [
  0x243f6a88, 0x85a308d3, 0x13198a2e, 0x03707344,
  0xa4093822, 0x299f31d0, 0x082efa98, 0xec4e6c89
]
```

These are the fractional digits of π (same as Blowfish).

2.3.1. Myszkowski permutation

Keyword: `MYSZKOWS` (8 letters, S repeated at positions 3 and 8).

Alphabetical order: K(5), M(1), O(6), S(3,8), W(7), Y(2), Z(4).

The byte permutation `MYZ_PERM[0..31]` is constructed as:

```
MYZ_PERM = [
   4,12,20,28,                 # K (column 4)
   0, 8,16,24,                 # M (column 0)
   5,13,21,29,                 # O (column 5)
   2, 7,10,15,18,23,26,31,     # S,S (columns 2 and 7, read row-wise)
   6,14,22,30,                 # W (column 6)
   1, 9,17,25,                 # Y (column 1)
   3,11,19,27                  # Z (column 3)
]
```

The inverse permutation `MYZ_INV` satisfies `MYZ_INV[MYZ_PERM[i]] = i`.

Design note. The Myszkowski choice is didactic (classical transposition with a keyword) and adds no cryptographic argument. A byte permutation optimized for diffusion (e.g., a structured permutation over the 4×8 byte grid) would strictly dominate it. We keep Myszkowski in this draft because it is the design under review; a future version may replace it.

2.4. Key schedule

2.4.1. Initialization

Input: key `K[0..63]` (64 bytes).

Initial state `s[0..15]` (16 words of 32 bits):

```
s[i] = big_endian_word(K[4i .. 4i+3])
```

2.4.2. Expansion

For `r = 0 .. 127`:

```
spread = (r + 1) * 0x9E3779B9  (mod 2^32)

# Round subkeys
for i in 0..7:
    rk[r][i] = s[i] ⊕ s[(i+5) mod 16] ⊕ s[(i+11) mod 16]
               ⊕ CONST[(r+i) mod 8] ⊕ spread

# For i = 0..3: fold fk into rk
for i in 0..3:
    fk = s[(i+8) mod 16] ⊕ CONST[(r+i+3) mod 8]
```

```
    rk[r][i] = rk[r][i] ⊕ fk

# State update (Feistel 8+8, bijective)
L = s[0..7]
R = s[8..15]
for i in 0..7:
    a = R[i]
    b = R[(i+1) mod 8]
    c = R[(i+3) mod 8]
    F = (rotl((a + b) mod 2^32, 7) ⊕ rotl(c, 13) ⊕ CONST[i] ⊕ spread)
    s[i]     = R[i]
    s[i + 8] = L[i] ⊕ F
```

2.4.3. Final whitening

After the 128 rounds:

```
for i in 0..7:
    rkFinal[i] = s[i] ⊕ s[(i+8) mod 16] ⊕ CONST[(128+i) mod 8]
```

2.5. Round function

Each round `r` transforms state `s[r]` into `s[r+1]` in four steps:

Step 1. AddRoundKey.

```
for i in 0..7:
    s[r][i] = s[r][i] ⊕ rk[r][i]
```

Step 2. Feistel cells. For `p = 0..3`:

```
L = s[r][p]
R = s[r][p + 4]
s_mid[p]     = R
s_mid[p + 4] = L ⊕ F(R)
```

where `F(x) = (rotl(x,1) ∧ rotl(x,8)) ⊕ rotl(x,2)`.

Step 3. Myszkowski transposition. Convert `s_mid[0..7]` to bytes (big-endian per word):

```
bytes[4i .. 4i+3] = big_endian(s_mid[i])
permuted[j] = bytes[MYZ_PERM[j]]  for j = 0..31
s_perm[i] = big_endian_word(permuted[4i .. 4i+3])
```

Step 4. Word rotation.

```
shift = (r mod 31) + 1
for i in 0..7:
    s[r+1][i] = rotl(s_perm[i], shift)
```

2.6. Encryption

```
s[0] = plaintext  (as 8 big-endian words)
for r = 0..127:
    apply round r (Section 2.5)
for i in 0..7:
    s[128][i] = s[128][i] ⊕ rkFinal[i]
ciphertext = s[128]  (as 32 big-endian bytes)
```

2.7. Decryption

Inverse of encryption:

```
s[128] = ciphertext
for i in 0..7:
    s[128][i] = s[128][i] ⊕ rkFinal[i]
for r = 127..0:
    apply inverse round r (inverse of Section 2.5, reverse order)
plaintext = s[0]
```

The inverse round consists of:

1. Inverse rotation: `rotl(word, 32 - shift)`.
2. Inverse transposition: `bytes[MYZ_PERM[j]] = permuted[j]`.
3. Inverse Feistel: `R = newL`, `L = newR ⊕ F(R)`.
4. XOR with `rk[r][i]`.

2.8. Mode of operation (CTR)

32-byte counter:

```
counter[0..15]  = nonce (16 bytes)
counter[16..31] = 0  (128 bits, big-endian)
```

For each 32-byte plaintext block:

```
keystream = MYSX_encrypt(counter)
ciphertext_block = plaintext_block ⊕ keystream
counter[16..31] += 1  (big-endian)
```

2.9. AEAD (encrypt-then-MAC)

Packet: `VERSION(1) ‖ NONCE(16) ‖ CT(n) ‖ TAG(32)`

MAC input: `VERSION ‖ NONCE ‖ AAD_LEN(4 BE) ‖ AAD ‖ CT_LEN(4 BE) ‖ CT`

TAG: `HMAC-SHA256(macKey, MAC input)`

Verification: the decryptor verifies the TAG before decrypting.

2.10. Key derivation

From a random secret (HKDF-SHA256).

```
encKey = HKDF-SHA256(secret, salt, info = context ‖ ":enc", L = 64)
macKey = HKDF-SHA256(secret, salt, info = context ‖ ":mac", L = 32)
```

From a password (PBKDF2-SHA256 + HKDF).

```
prk    = PBKDF2-SHA256(password, salt, iterations = 600000, L = 32)
encKey = HKDF-SHA256(prk, salt, info = context ‖ ":enc", L = 64)
macKey = HKDF-SHA256(prk, salt, info = context ‖ ":mac", L = 32)
```

2.11. Test vectors

See `KAT.json` (public). Two digests are documented in `README.md`:

- `KAT.json` public digest: SHA-256 over the full JSON (block + AEAD + KDF). This is the artifact a third-party implementation should reproduce.
- `KNOWN_KAT_DIGEST` (internal): SHA-256 over 16 blocks with `key[i] = i`. Used as fast regression in `mysx.test.js`. It is strictly a subset of the public KAT, not a different KAT.

A third-party implementation should target the **public digest**.

3. Differential cryptanalysis

3.1. Model

Tool: `misc/mysx_milp.py` (PuLP + CBC/HiGHS). AND model: `a = Δx ∨ Δy`, which counts a cost-1 event whenever at least one input is active. For an AND with independent inputs, this is the exact per-bit cost. The real approximations in the model are:

- The Markov-independence assumption does not hold: the two rotated inputs of F come from the same word `x`.
- We ran a single input difference at r=12–14 (w2b30). Multi-bit input differences may yield lower weights.
- No run varying the rotation offset (index of the first round mod 31) is reported. The linear analysis of §4.10 shows that this offset changes the best achievable weight by an order of magnitude (at R=14, backward: 5 at offset 20 vs 61 at offset 6). The differential bounds below are therefore not known to be representative, and the extrapolation to r≈15 is likely optimistic until the offset is varied.
- `gapRel=0.2` and `timeLimit=300` return feasible solutions (upper bounds), not certified optima.
- The model captures single trails, not differentials (clusters of trails with the same output difference).

3.2. Observed upper bounds

Rounds	Input	Weight (upper bound)	Single-trail prob.
5	w1b24	2	2^{-2}
6	w1b24	6	2^{-6}
7	w2b30	10	2^{-10}
8	w2b30	18	2^{-18}
12	w2b30	68	2^{-68}
14	w2b30	114	2^{-114}

Extrapolation: the 128-bit threshold is reached at r≈15. **This is an extrapolation, not a proof.** It tells us what the model produces for the input we tried; it does not establish the best trail over all inputs or offsets, nor that clustering does not lower the effective security.

Comparison with the linear direction. At 14 rounds the differential upper bound is 114 ($\approx 2^{114}$ pairs), whereas the linear side now has a **verified distinguisher on a 14-round window** (offset-20 backward trail covering rounds 100–113, weight 5, $\approx 2^{10}$ texts, isolated hull; §4.10.3). Even allowing for the different thresholds (§1.1) and for the fact that the differential run used a single input difference with a 20% solver gap, the linear direction is the weaker one by a wide margin and is where analysis effort should concentrate. Part of that gap is probably not intrinsic: the linear figure is the minimum over all 31 rotation offsets, the differential figure is for one input at one position. A differential analysis with inputs placed in the free windows and over all offsets (the analogue of the weight-0 linear chains and of §4.10) has not been done.

Direct verification at r=4 (w=32). The r=4 trail (it is not a row of the table above: Δin = {w0b0}, MILP weight 8, Δout = {w0b4, w1b10, w1b16, w1b30, w6b4}) was verified with 8 fixed keys × 2^{17} pairs each, using the full 128-round cipher with the final whitening disabled (to isolate the round function). Observed hit rate: 7.802×10^{-3} over 1,048,576 pairs. Expected from MILP: $2^{-8} = 3.906 \times 10^{-3}$. **Ratio: 2.00×.**

The trail exists at r=4. The single-trail probability is a factor 2 higher than the MILP predicts, consistent with the clustering factor documented in §3.3. **The MILP model is correct within a factor of 2 at this round count.**

3.3. Empirical check (r=4, w0b0)

As described above, the direct verification gives a ratio of 2.00× between the observed probability and the MILP single-trail prediction. Two interpretations are consistent with the data:

- **Differential clustering** (more likely): multiple trails with the same output difference Δout contribute to the observed probability, so the effective probability is higher than the single-trail probability.
- **Model systematic overestimate of the cost** (less likely): the AND model $a = \Delta x \vee \Delta y$ charges one bit too much at r=4. This would imply the true weight is 7, not 8. (Because the observed probability is *higher* than predicted, the model cannot be *under*estimating the cost; an underestimate would give a probability lower than predicted and a true weight of 9.)

Without a full clustering analysis (summing over all trails with the same Δin, Δout), we cannot distinguish these two. The practical implication is the same: the effective weight of the r=4 trail is 7, not 8, so the MILP weight is the pessimistic side by one bit.

What this does to the extrapolation. A *constant* factor 2 is one bit of weight; at the observed 8–23 bits of weight per round it moves the "128-bit threshold at r≈15" by a small fraction of a round, not by 2–3 rounds. Only a factor that grows with the weight (for example the ratio 7/8 applied to the whole trail) would matter: it would move the crossing to roughly r≈16–17. Either way the margin on this input is large, and the extrapolation should be read as approximate and, see §3.1, as specific to the input and offset tried.

3.4. What this section does not establish

- A lower bound on the best single-trail weight (requires gap=0 and multi-bit inputs).
- An exact clustering analysis (requires modeling all trails with the same Δout).
- Resistance to related-key differentials.
- That the factor-2 clustering is constant across all round counts.
- The effect of the rotation offset (first round mod 31) on the best differential weight, nor any differential inputs placed in the free windows.

4. Linear cryptanalysis

4.1. Overview

An earlier version of this paper (§4 of the v0.0 draft) reported only negative results: three approximate linear MILP models (`misc/mysx_linear_milp.py`) all produced trails that failed empirical verification at w=32. That negative result has now been explained. The models used per-cell AND activity (`act = α ∨ β ∨ γ`) as a proxy for correlation, which is not the correct structure of F's LAT. A correct model exists in closed form and is implemented in `cryptanalysis/models/linear/mysx_linear_model.py` (with detailed documentation in `cryptanalysis/models/linear/LINEAR_MODEL.md` and raw data in `cryptanalysis/models/linear/linear_model_data.json`). The model is machine-checked against the real cipher by `python3 mysx_linear_model.py selftest`.

Result in one line. The LAT of F is exactly `corr(b,g) = ±2^{−h(b)}`, where `h(b)` is half the GF(2)-rank of a 32-cycle subgraph determined by `b`; there is a free choice of `g` among `4^{h(b)}` values with sign determined by a quadratic form. Mask propagation is bit-exact; every trail carries an exact key-dependent sign. Verified consequences: `|corr|=1` chains over up to 9 consecutive rounds (three keys), and exact linear-hull predictions at r=10,11,12 that match the real cipher within sampling error ($|z| < 0.4$). §4.10 extends the search for cheap trails to 128 rounds, all rotation offsets, and **both directions** (forward and backward), and shows that single-trail weights must be read together with hull effects. For the offset-20 backward trail at R=14 (weight 5, covering rounds 100–113), the hull check is complete: the trail is isolated and its

empirical correlation matches 2^{-5} (§4.10.3). This is a verified linear distinguisher on a 14-round window of the cipher.

4.2. Exact LAT of F

For output mask `b` and input mask `g`,

```
b.F(x) ⊕ g.x  =  Q_b(x) ⊕ l.x ,   l = rotr(b,2) ⊕ g ,
Q_b(x) = ⊕_j b_{j+1} x_j x_{j−7} .
```

`Q_b` is a quadratic form whose graph `G_b` is a subgraph of the 32-cycle `j — j−7` (edge `(j, j−7)` present iff `b_{j+1}=1`; a single cycle because `gcd(7,32)=1`). Let `A_b` be the GF(2) adjacency matrix of `G_b`. Then:

- `corr(b,g) ≠ 0` iff `l = A_b y` for some `y`. There are exactly `2^{rank(A_b)} = 4^{h(b)}` valid `g`.
- For each valid `g`, `corr(b,g) = (−1)^{Q_b(y)} · c_0(b)` with `|c_0(b)| = 2^{−h(b)}`, where `h(b) = rank(A_b)/2`.
- `h(b)` equals the maximum matching = minimum vertex cover of `G_b` (König). The only graph that is not a forest is `b = 0xFFFFFFFF` (full cycle), where `rank = 30` and `h = 15`.
- Parseval holds: `Σ_g corr(b,g)² = 1`.

This is **exact** and **machine-checked** against brute-force LAT of F for random `b` of Hamming weight up to 8 (`selftest`).

Cost data.

- Single-bit `b`: `h = 1`, four valid `g`, `|corr| = 1/2`.
- Two-bit `b` (496 masks): histogram `{2: 464, 1: 32}` (the 32 pairs at bit distance 7 are adjacent edges).
- Uniform random 32-bit `b` (16384 samples): histogram of `h` peaks at `h = 11` (4130 samples); full histogram in `linear_model_data.json`.
- Densest masks per unit of weight (bits `(1−7t) mod 32`, `t < 2m`): every extra 2 consecutive edges costs 1; cost per active mask bit approaches 1/2 in the best case, never below 15/32.

4.3. Exact mask propagation (forward and backward)

Forward step. For a round `r`, with input mask `m` on `s[r]` and cells `p = 0..3` (`L = s[r][p]`, `R = s[r][p+4]`):

```
b_p     = m[p]                              (only words 0..3 decide the cost)
g_p   ∈ valid_gammas(b_p)          (4^{h(b_p)} choices, each |corr| = 2^{−h(b_p)})
n[p+4]  = m[p] ;  n[p]  = m[p+4] ⊕ g_p
m_next  = lin_fwd(n, r)            (byte transposition, then rotl by (r mod 31)+1)
w_r     = Σ_p h(m[p])
sign    = (Π_p sign_p) · (−1)^{m · rk[r]}
```

Backward step. For a round `r`, given `m_next` on `s[r+1]`:

```
n     = lin_inv(m_next, r)         (inverse rotation, then inverse transposition)
```

```
b_p    = n[p+4]                      (this is the output mask of F at cell p)
m[p]    = n[p+4]                         (cost h(n[p+4]) at this round)
m[p+4]  = n[p] ⊕ g_p ,  g_p ∈ valid_gammas(b_p)
```

The `selftest` verifies that every forward edge is a backward edge with the same cost (325 edges checked), so the two searches enumerate the same set of trails. Trails are always stored in forward order (mask on `s[first]` … mask on `s[last+1]`).

Correlation of a trail: `Π_r 2^{−w_r} · (−1)^{Σ_r m_r·rk[r]} · Π_p sign_p`. The linear hull at fixed key is the exact sum over all trails with the same `(m_0, m_R)`; `propagate()` computes it by correlation-vector propagation (pruned by weight). ELP mode (`propagate(..., elp=True)`) gives the key-independent potential under the (incorrect for a deterministic schedule) independent-round-key assumption.

Free-chain structure. A round has weight 0 iff words 0..3 of its input mask are zero; the round then acts as a pure bit permutation and the tracked bit never meets `F`. Since the rotation schedule has period 31, there are only 31 distinct windows. The exact free-chain table (all 128 single-bit masks in words 4..7, 31 windows) is in `LINEAR_MODEL.md` §4 and `linear_model_data.json`. Headline: the longest weight-0 chain is **9 consecutive rounds**, at windows `r ≡ 12 (mod 31)` and `r ≡ 27 (mod 31)`.

4.4. Verified `|corr| = 1` chains

Each of the 9-round chains has been checked against the real cipher (fixed random keys, 2^{21} plaintexts, reduced rounds, final whitening disabled to isolate the round function). Representative results:

window r0	rounds	start bit	end bit	key seed	predicted sign	empirical corr	σ
12	9	s[12].174	s[21].79	1	+1	+1.00000	6.9e-04
12	9	s[12].174	s[21].79	2	−1	−1.00000	6.9e-04
12	9	s[12].174	s[21].79	3	+1	+1.00000	6.9e-04
27	9	s[27].146	s[36].55	1	+1	+1.00000	6.9e-04
27	9	s[27].146	s[36].55	2	+1	+1.00000	6.9e-04
27	9	s[27].146	s[36].55	3	+1	+1.00000	6.9e-04
0	5	s[0].208	s[5].55	1	−1	−1.00000	6.9e-04
0	5	s[0].208	s[5].55	2	−1	−1.00000	6.9e-04
0	5	s[0].208	s[5].55	3	−1	−1.00000	6.9e-04

The magnitude is 1 for every key. Only the sign depends on the key, through `Σ_r m_r · rk[r]`. The model predicts that sign exactly. Per- round masks and complete details are in `linear_model_data.json` (`verified_chains`).

The backward search finds the same 9-round weight-0 chains at the corresponding end residues (20 and 4 mod 31). The `selftest` verifies `|corr|=1` on the real cipher for both, and that the forward and backward chains are the same trails (stored in forward order).

4.5. Best single trails beyond the free chain (upper bounds)

Layered beam search with the exact rules, starts = all 256 single-bit masks, beam 1200, cells with `h ≤ 2` expanded. Each reported weight is an **achievable single-trail weight**, i.e. an upper bound on the best weight, not a minimum.

rounds R	1–9	10	11	12	13	14
best weight, window r0=12	0	1	2	3	7	12
best weight, window r0=27	0	1	3	3	6	11

This table covers only the two windows with the longest free chain. §4.10 searches all 31 offsets in both directions and shows that the windows 12 and 27 are the best only up to R=12; from R=13 on, other offsets (notably r0=23 forward and r0=20 backward) give lower single-trail weights.

Note that this is *the opposite* of a linear-distinguisher risk at low rounds: at the 9-round mark we still have weight 0 (correlation ±1 in absolute value). The reader should not confuse this with an attack on the full cipher — see §4.7.

4.6. Hull verification (r0=12, exact signed hulls)

The single-trail numbers above are inputs to a hull computation. For three trails at r0 = 12 with weights 1, 2, 3, the model produces an exact signed prediction (fixed key, sum over all trails of weight ≤ w+4 with the same end masks) and the real cipher produces an empirical correlation over 2^{22} plaintexts.

window r0	R	best-trail w	hull masks kept	predicted corr	empirical corr	σ	z
12	10	1	4	+0.500000	+0.499997	4.9e-04	−0.01
12	11	2	40	−0.250000	−0.250008	4.9e-04	−0.02
12	12	3	3152	−0.125000	−0.124813	4.9e-04	+0.38

Three independent predictions, three matches, $|z| < 0.4$ in every case. The hull sum is not an approximation here; it is the exact polynomial expansion of the trail system in the model, and it is what the real cipher produces within sampling error.

§4.10.3 extends this check to the saved R=14 trails from the beam search. The r0=23 forward trail cancels from R=8; the r0=24 forward trail cancels from R=8; the r0=12 forward trail is exact up to R=12; the offset-20 backward trail is exact from R=1 to R=14. Hull effects are not a uniform correction.

4.7. What this establishes and what it does not

Established (machine-checked by `selftest`):

- Exact LAT of F (valid set, magnitude, sign) against brute force.

- Exact per-round mask rule and key-sign tracking, in both directions (forward and backward edges match, 325 edges checked).
- Existence of key-independent-magnitude `|corr| = 1` single-bit relations over up to 9 consecutive rounds; found by both forward and backward searches at the corresponding residues.
- Exact signed linear-hull predictions (not approximations) at r=10, 11, 12 against the real cipher.
- The `mysx.prod.js` Python port reproduces `KNOWN_KAT_DIGEST`, so all claims are made against the actual reference implementation.
- (§4.10) The beam-search tool reproduces the §4.5 table exactly, its per-round masks agree with the real cipher wherever a trail is its own hull, and the `selftest` runs in both directions.
- (§4.10.3) The offset-20 backward trail at R=14 (weight 5, rounds 100–113) is isolated in its hull (`hull_count = 1` trail) and its empirical correlation matches 2^{-5} on the real cipher. This is a **verified linear distinguisher on a 14-round window**. It is five cheap rounds followed by a 9-round weight-0 chain, so it is the free chain of this section plus a short cheap prefix, not new evidence about long trails.

Not established:

- A **lower bound** on the best trail weight over 128 rounds. Beam search gives upper bounds from a specific start distribution.
- An exact 128-round hull. Even the pruned `propagate()` will not scale; a certified bound would need a SAT/MILP encoding.
- A hull-aware correlation for the trails found in §4.10 beyond R=14. §4.10.3 shows that single-trail weight can over- or under-estimate the real correlation (cancellation was observed for some trails and isolation for others).
- The cost of a key-recovery attack.
- Behaviour of multi-bit starts at larger R beyond the pairs47 and rand:5000:4 runs (which did not improve on single-bit; rand:5000:4 is too small to be conclusive for structured multi-bit masks).
- Anything about the last-round / whitening structure, related-key linear behaviour, or side channels.

The weight-0 chains are not an attack. A 9-round `|corr| = 1` relation sits in a 128-round cipher. Standard linear cryptanalysis extrapolates a weight over the full length; here the trail weight over rounds 0..8 is 0 and the weight over rounds 9..127 is unknown. The result is best read as: *the exact machinery is in place and it agrees with the real cipher on small round counts*, not as *MYSX is broken*. The 128-round picture remains open, although §4.10 shows that the cheap region ends at roughly 20 rounds in the single-trail model.

4.8. Previous incorrect models and why they failed

For the record, the v0.0 draft reported three approximate models, all with negative empirical verification. They used per-cell AND activity as a proxy for correlation, which is not the correct structure of F. The correct structure is the rank of the graph `A_b` above. The previous models produced trails with predicted `|corr| ≈ 2^{-8}` at

r=4 but no detectable signal — consistent with the fact that those trails have zero true correlation in the exact model.

An instructive detail from the old work remains relevant to *any* future linear work on MYSX: the empirical verification harness must use **fixed keys** per sample. With random keys per plaintext, the sign of every linear trail is key-dependent and the observed correlation of a genuinely nonzero trail collapses to zero. See `misc/validate_lineal_harness.js`, which validates the harness against a hand-computed `w=8` case (observed correlation 0.4993 vs 0.5).

4.9. Integral / division property (preliminary, limited)

Tool: `misc/mysx_division_property_v2.py` with standard division- property rules (COPY: `k_a = Σ k_copies`; XOR: `k_c = k_a + k_b` with `(1,1)` invalid; AND: `k_c = k_a OR k_b`). The XOR and OR variants were found to be equivalent for all input patterns tested (the `(1,1)` constraint never fired), so the reported results do not depend on the choice between them.

Results.

Input	Active bits	Integral size	Rounds with balanced bits	Saturation
bit 0 of word 0	1	2	10	R = 11
word 0	32	2^{32}	6	R = 7
words 0 and 4	64	2^{64}	5	R = 6

The **integral distinguishers** are the last two rows: 6 rounds with a 2^{32}-text integral, 5 rounds with a 2^{64}-text integral. These cover 4.7% and 3.9% of the 128 rounds, leaving 122 and 123 rounds without an integral distinguisher in the tested pattern.

The first row (1 active bit, 2-text integral) is **not** a meaningful integral distinguisher: with 2 texts, "balanced output bit" is equivalent to "output bit independent of the input bit". It is reported as a **single-bit dependency propagation** result — it measures how many rounds are needed for the active bit to reach all 256 output bits (10) — but no integral security conclusion is drawn from it.

Empirical verification. The 6-round distinguisher (word 0 active) and the 10-round single-bit dependency were verified against the real cipher (fixed random key, all non-active input bits random) over 20,000 samples each. The balanced bits predicted by the model never flipped; control bits (predicted not balanced) flipped with probabilities 6.5%–87%. The 5-round distinguisher (words 0,4 active) was not verified empirically (its integral has 2^{64} texts; a random- sample approximation with 20,000 samples is not meaningful).

Comparison with Simon (with care). For Simon32/64, Todo and Morii (FSE 2016) prove a 14-round integral distinguisher with the conventional bit-based division property (31 active input bits), and a 15-round one using three subsets. 14 rounds is 44% of Simon32's 32 rounds (47% for 15 rounds). In **relative terms** MYSX resists integral distinguishers better (4.7% vs 44%), which is consistent with MYSX's more

aggressive diffusion (4 parallel Feistel cells + transposition). In **absolute terms** Simon's distinguisher reaches further (14 vs 6 rounds) even though MYSX has 4× more rounds. The two are not directly comparable without normalizing by total rounds, and the MYSX analysis used only the conventional bit-based rules at r0=0. (The earlier text of this paragraph quoted an 11-round Simon32 distinguisher, 34%; that figure was wrong.)

What this establishes and what it does not.

Established:

- MYSX has integral distinguishers covering 5–6 rounds with multi-bit inputs (2^{32} and 2^{64} texts).
- A single active input bit stops influencing all 256 output bits after 10 rounds.
- The standard division-property rules with the `(1,1)`-invalid XOR constraint produce the same results as the OR-conservative rule for the tested inputs.

Not established:

- Integral distinguishers beyond 6 rounds for multi-bit inputs.
- Behaviour for r0 ≠ 0. Only r0 = 0 (the first round window) was tested. The rotation schedule has period 31, and windows r0 = 12 and r0 = 27 are where the linear chains are longest; they may also give longer integral distinguishers, or shorter. Not explored.
- Full-block inputs (255 active bits).
- The three-subset division property (introduced by Todo–Morii, FSE 2016; variant without unknown subset: Hao et al., EUROCRYPT 2020) and traceability-enhanced variants (Wang et al., 2023; citation to be verified), which may extend the distinguishers.
- Any key-recovery attack based on these distinguishers.

Historical note. The XOR rule of Todo–Morii as commonly written in the literature (`k_c = k_a XOR k_b`) is not sound when the two XOR operands share variables. In MYSX's `F = C ⊕ D`, both `C` and `D` depend on `R`, so that rule overestimates balanced bits via spurious cancellations. The first implementation of this analysis used that rule and produced oscillating, non-converging numbers (127 → 117 → 152 → ...). Those numbers are an artifact and are not reported here. The results above use the standard `(1,1)`-invalid rule, which is sound.

4.10. Beam search over the exact linear model, 128 rounds, both directions, all offsets

4.10.1. Method

Tool: `misc/mysx_linear_beam.py` (self-contained: it re-implements the exact LAT of F, the exact mask propagation of §4.3 in both directions and, for its `selftest`, the real cipher). It searches for achievable single-trail weights with `w = Σ_r Σ_p h(m_r[p])`.

Two directions.

- **Forward** (`--direction forward`, default): the offset `o` is the index of the *first* round mod 31. Starts are masks on `s[o]`, the trail covers rounds `o .. o+R-1`,

and the trail must fit inside the cipher (`o + R ≤ 128`).

- **Backward** (`--direction backward`): the offset `o` is the index of the *last* round mod 31. Starts are masks on `s[e+1]` with `e = o + 31·⌊(127−o)/31⌋`, the trail covers rounds `e-R+1 .. e`, and the trail must fit inside the cipher (`e + 1 ≥ R`, i.e. `realizable_R = e + 1 ≥ R`). The propagation uses the inverse round rule of §4.3.

Merge. `python3 mysx_linear_beam.py merge a.json b.json c.json` combines any number of result files (from either direction) and reports the minimum weight per R across all offsets and both directions. The combined table of §4.10.2 is the output of `merge beam_full.json beam_pairs47.json beam_bwd.json`.

Parameters of the reported runs:

Parameter	Value
Rotation offsets	all 31 (0..30)
Rounds	up to 128
Start masks	the 256 single-bit masks of the relevant state
Beam width	2000 states per layer, ranked by (weight, Hamming weight), deduplicated by mask
Per-cell limit	`h(b) ≤ 16`; `b = 0xFFFFFFFF` (full cycle, special case of §4.2) is excluded
Valid `g` per cell	the 3 sparsest elements of the valid set (`--g-keep 3`)
Reported value	suffix minimum `ub(R) = min_{R' ≥ R} best(R')`, valid because the best weight is non-decreasing in R
Forward run	`beam_full.json`, `beam_full.log`: 8 processes, 1972 s
Backward run	`beam_bwd.json`, `beam_bwd.log`: 8 processes, 1512 s
Multi-bit run	`beam_pairs47.json`: `--starts pairs47`, did not improve on single-bit

Every exclusion or truncation above only *restricts* the search, so all reported weights remain achievable single-trail weights (upper bounds on the best weight). A larger beam, `--g-keep 0` and multi-bit starts can only lower them.

Validation of the tool, before the runs:

- `selftest` passes in both directions: (1) the closed-form LAT of F against sampling, (2) one round of the model against the real round function with random keys, (3) the 9-round weight-0 chains at r0 = 12 and r0 = 27 found by the forward search and confirmed with `|corr| = 1` on the real cipher (256 random states per chain), (4) every forward edge is a backward edge with the same cost (325 edges checked), (5) the 9-round weight-0 chains are found backward at end residues 20 and 4 with `|corr| = 1` on the real cipher, (6) a backward R=14 trail with weight 21 is a valid forward trail with equal weight.

- `reproduce` returns exactly the §4.5 table for r0 = 12 and r0 = 27 (R ≤ 14, beam 1200, `h ≤ 2`).

4.10.2. Results

Best achievable single-trail weight per number of rounds, minimum over offsets and both directions, subject to the trail fitting inside the 128-round cipher (forward: `r0 + R ≤ 128`; backward: `realizable_R = e + 1 ≥ R`):

R	best weight	offset	direction	≈ texts 2^{2w}
9	0	4	backward	2^{0}
10	1	4	backward	2^{2}
12	2	20	backward	2^{4}
14	5	20	backward	2^{10}
16	8	23	forward	2^{16}
18	24	20	backward	2^{48}
20	38	20	backward	2^{76}
22	63	23	forward	2^{126}
24	88	3	backward	2^{176}
32	204	3	backward	2^{408}
40	361	20	backward	2^{722}
48	485	20	backward	2^{970}
56	630	27	forward	2^{1260}
64	786	27	forward	2^{1572}
80	1081	9	forward	2^{2162}
96	1336	21	backward	2^{2672}
112	1671	21	backward	2^{3342}
128	2248	3	backward	2^{4496}

Largest R with an achievable single trail of weight < 32 (≈2^{64} texts): **19**. Largest R with weight < 64 (≈2^{128} texts): **22** (weight 63 at R=22, offset 23 forward; 49 at R=21). This is a **tie with the threshold**: at beam 1000 the same cell gives 64/71, so the defensible statement is “≈20–22 rounds”, not “22”.

Direction alternates with R. Among the tabulated round counts the minimum is reached by a backward trail at R=9, 10, 12, 14 (tied with offset 23 forward), 18, 20, 24, 32, 40, 48, 96, 112 and 128, and by a forward trail at R=16, 22, 56, 64 and 80. The two directions align with different sets of rotation windows, so neither is uniformly better. The combined table is the right object; do not compare the forward-only and backward-only tables row by row.

R=128 is a real trail. In forward mode only r0=0 fits a 128-round trail (`o + R = 128`). In backward mode, offset 3 has `e = 3 + 31·⌊124/31⌋ = 127`, `realizable_R = 128`, and the trail covers rounds 0..127. The combined R=128 = 2248 at offset 3 backward is a **real trail inside the cipher**, not a virtual extension. Offsets with `realizable_R < R` are excluded from the minimum (the tool checks this explicitly). The `v0.1-draft` statement that “for r0>0 the value at R=128 is a virtual extension” was forward-specific; it does not hold for backward offset 3.

Per-offset detail at R=14 (single-bit starts, beam 2000, merged). For compactness, the backward values are shown (they include the best cell of the whole table, offset 20 = 5); the forward values are in `beam_full.json` and the merged per-offset arrays are in the output of `merge`.

offset (backward, last round mod 31)	0	1	2	3	4	5	6	7	8	9	10
weight	17	11	11	8	11	36	61	46	34	47	43

offset	11	12	13	14	15	16	17	18	19	20	21
weight	24	53	42	38	31	24	16	12	8	**5**	21

offset	22	23	24	25	26	27	28	29	30
weight	25	21	13	19	41	36	36	28	26

The best cell at R=14 is offset 20 backward (weight 5). The next best are offsets 3 and 19 backward (weight 8). The forward per-offset table is available in `beam_full.json`; the best forward cell at R=14 is offset 23 (weight 5), tied with offset 20 backward. The spread (5 to 61 across offsets for the same R) is the evidence that the rotation offset matters a great deal, which is also why the differential analysis of §3 needs to be repeated over offsets.

Growth. Between R=22 and R=112 the best weight grows by roughly 12–21 per round (12.5 for 22→24, 14.5 for 24→32, 19.6 for 32→40, 15.5 for 40→48, 18.1 for 48→56, 19.5 for 56→64, 18.4 for 64→80, 15.9 for 80→96, 20.9 for 96→112). The last step, 112→128, is ≈36 per round (1671 → 2248). That step is inflated by the fitting rule: only offset 3 backward fits a full 128-round trail, so the R=128 value is not a minimum over all 31 offsets as the others are. The average between R=48 and R=128 is ≈22 per round. The weight never settles at a low per-round cost. Note also that single-trail weights above ≈256 say nothing about the cipher: with a 256-bit block the sum over the many trails of a hull is what matters there, and no hull-aware statement is made beyond R≈14.

4.10.3. Hull check of the trails found

A single-trail weight `w` predicts `|corr| = 2^{-w}` only when the trail is alone in its hull. The beam trails were therefore checked against the real cipher (fixed key, final whitening disabled, 2^{17}–2^{18} random states), prefix by prefix, using the stored per-round masks. Script: `misc/verify_beam_trail.py`; trail counting: `misc/hull_count.py`.

Trail of offset 20 backward, R=14, w=5 (covers rounds 100..113; `first_round[14] = 100`). Key seed 1, 2^{17} states (σ = 0.0028):

R	trail weight w	predicted 2^{−w}	empirical corr
1	1	0.5000	−0.4947
2	3	0.1250	−0.1206
3	4	0.0625	−0.0601
4	4	0.0625	+0.0601
5	5	0.0312	+0.0279
6	5	0.0312	−0.0279
7	5	0.0312	−0.0279
8	5	0.0312	−0.0279
9	5	0.0312	−0.0279
10	5	0.0312	+0.0279
11	5	0.0312	+0.0279
12	5	0.0312	−0.0279
13	5	0.0312	+0.0279
14	5	0.0312	−0.0279

`hull_count` on the saved end mask returns **exactly one trail** at the minimum weight for every R from 1 to 14 (layer sizes 4, 64, 84, 10, 4, 2, 1, 1, ..., 1; the layer collapses to a single trail from R=7). The predicted and empirical magnitudes agree within σ at every R. There is **no cancellation** for this end mask. **This is a verified linear distinguisher on a 14-round window (rounds 100–113).**

How to read it. The weight reaches 5 at R=5 and stays there up to R=14: the trail is five cheap rounds followed by a 9-round weight-0 chain (the free chain of §4.3). The rows R=5...14 are therefore the same signal observed after more free rounds, not ten independent confirmations. The trail lies at rounds 100–113, so it is a property of that window of the cipher; it is not a distinguisher for a 14-round version of the cipher starting at round 0, and it is not an attack on the 128-round cipher.

Trail of r0=23 forward, R=14, w=5 (start bit 197 of `s[23]`; end masks bits {46, 66, 84, 196} of `s[37]`), key seed 1, 2^{17} states (σ = 0.0028):

R	trail weight w	predicted 2^{−w}	empirical corr
1–4	0	1	±1.0000
5–7	1	0.5000	−0.5014
8–10	2	0.2500	±0.0010
11–13	3	0.1250	±0.0012
14	5	0.0313	−0.0001

The model is exact up to R=7. At R=8 the empirical correlation is zero instead of ±1/4. An exact enumeration of all trails with weight ≤ 4 from the same start mask shows why: **two distinct trails of weight 2 reach the same end masks at R=8, 9 and 10** (and one at R=7), so their contributions cancel. Two independent keys at

R=14 (2^{18} states each, σ = 0.002) give −0.0004 and +0.00001 against a single-trail prediction of 0.031. For this trail the single-trail weight is therefore not a usable estimate of the correlation.

Other offsets, R=14 trails, one key (seed 1), 2^{16} states (σ = 0.0039):

r0	direction	R	single-trail 2^{−w}	empirical corr	relation
12	forward	10	0.5000	+0.4969	equal
12	forward	12	0.1250	−0.1161	equal (sign key-dependent)
27	forward	10	0.1250	+0.2503	×2
27	forward	12	0.0156	+0.0369	×2.4
26	forward	10	0.1250	+0.2412	×2
26	forward	12	0.0625	+0.1218	×2
24	forward	10	0.1250	+0.0049	cancelled
24	forward	12	0.1250	−0.0049	cancelled

At R=14 the predicted correlations (2^{-7} to 2^{-12}) are below the resolution of this test (σ = 0.0039) for these four forward trails, so no statement is made at R=14 for them. The offset-20 backward trail (previous table) is measurable at R=14 because its weight stays at 5 from R=5 to R=14, so the test at R=14 is the same signal as at R=5 (where σ is adequate).

Reading. Across the trails checked, the real correlation equals the single-trail value (offset 20 backward at R=14; r0=12 forward up to R=12), is about twice as large (r0=26, r0=27 forward; a second trail adds constructively), or is exactly cancelled (r0=23 forward from R=8; r0=24 forward). The observed gain is at most ≈2.4× (about 1.3 bits); the observed loss can be total. Hull effects are therefore not a uniform correction, and they are not always destructive: some trails survive intact and are genuine distinguishers of their window.

4.10.4. What §4.10 establishes and what it does not

Established:

- In the single-trail model, and within the searched family (single-bit starts, beam 2000, 3 sparsest g per cell, both directions), no trail with weight < 64 (≈2^{128} texts) was found for more than 22 rounds, and none with weight < 32 for more than 19 rounds. The threshold at 22 is a tie (weight 63): "≈20–22" is the defensible reading.
- The best weight grows by roughly 12–21 per round from R≈22 to R=112 (≈22 on average from R=48 to R=128, inflated by the last step).
- The cheapest trails alternate direction (see §4.10.2 for the exact R values). The best cell at R=14 is offset-20 backward (weight 5), tied with offset-23 forward.
- The tool agrees with the real cipher wherever a trail is alone in its hull (offset-20 backward up to R=14; r0=12 forward up to R=12; r0=23 forward up to R=7).
- Single-trail weight is not a reliable proxy for correlation in general: exact cancellation was observed (r0=23 and r0=24 forward).

- The offset-20 backward trail at R=14 is isolated in its hull (`hull_count` = 1 trail) and its empirical correlation matches 2^{-5} on the real cipher. **This is a verified linear distinguisher on a 14-round window (rounds 100–113).**
- Multi-bit starts (pairs47, rand:5000:4) did not improve on single-bit starts at any R. pairs47 is a small structured subset; rand:5000:4 covers 5000 masks out of 2^{256} and is not conclusive for structured multi-bit masks.

Not established:

- A **distinguisher length beyond R=14**. R=14 is established for the offset-20 backward trail. R≥16 requires a hull-aware correlation for the chosen end masks; the trails of §4.10.2 at R=16–22 were not checked at the weights where this matters. The next step is the exact signed hull (`propagate()`) for the best end-mask pairs at R=16–22.
- A **lower bound**. The beam gives achievable weights only. The truncations (`g-keep 3`, beam, single-bit starts, exclusion of `b = 0xFFFFFFFF`) may hide cheaper trails.
- That hull gain stays small at larger R. At R=24 (w=88) a gain of $\approx 2^{24}$ in correlation ($88 - 64$) would be needed to bring the data cost down to the 2^{128} budget; the gains observed so far are ≤ 2.4× but were measured only up to R=14.
- That multi-bit starts do not help at all. The tested samples (pairs47, rand:5000:4) do not improve on single-bit, but the random sample is too small to rule out structured multi-bit masks that might. The honest statement is "the multi-bit starts tested do not improve on single-bit".
- Key recovery from any hull, differential analogues, and the effect of the final whitening and of key-recovery rounds around a distinguisher.

5. Related-key cryptanalysis

Tools: `misc/keyschedule_test.js` (basic) + `misc/keyschedule_test_extended.js` (22 tests). Coverage: exhaustive weight-1 (512 bits), exhaustive weight-2 (130,816 pairs), weight-3 "same Δ" (1,680 trials), beam search up to weight-6 (beam 50), 5 base keys, structured patterns, cycles, weak keys.

Results (summary):

- No persistent `Δrk[r]=0` for r≥1 in any test.
- Beam search score: 139 (ideal 128).
- No weak keys, cycles, or fixed points.
- Slow diffusion at r=1–4 (expected: one Feistel round per cipher iteration).
- Worst bit (minimum weight at r=1): bit 160 (word 5, bit 0), 1-bit difference. Consistent across 5 base keys (structural, not noise).

Limitations. These are heuristic tests, not a proof. We did not run a targeted MILP/SAT related-key model (~650k variables); adaptive related keys, biclique attacks, and algebraic attacks on the key schedule are not covered. **MYSX's related-key security is unknown.**

Round-key recovery. The only non-linear operation of the key schedule is the 8 modular additions per step; everything else is XOR and rotation. The schedule is

bijective, so any 16 consecutive state words determine the master key. How many round keys must be known for a SAT solver to recover the master key (an estimate by counting: about three consecutive `rk[r]`) has not been measured. This number bounds what a partial key-recovery attack on a few rounds can lead to.

6. Generic attacks

Attack	Complexity	Verdict
Classical brute force	2^{512} operations (uniformly random 512-bit key)	Infeasible
Grover (quantum)	2^{256} iterations (uniformly random 512-bit key)	Infeasible with foreseeable technology
Block collisions (birthday)	2^{128} blocks	Infeasible

Note on Grover. With a uniformly random 512-bit key, Grover leaves 256 bits of effective security. With 1,000,000 logical qubits at 1 ns/iteration, the time would be $\sim 10^{60}$ years. This is a theoretical estimate; no post-quantum analysis of the cipher beyond Grover has been done.

Note on the 512-bit key. The key-derivation functions of Section 2.10 (and therefore every key produced by `deriveAeadKeys` and `deriveAeadKeysFromPassword`) start from HKDF-Extract, whose output is a 256-bit pseudorandom key (HMAC-SHA256). HKDF-Expand can output 64 bytes, but they are a deterministic function of that 256-bit value, so the entropy of a derived 512-bit `encKey` is at most 256 bits. If the master secret has 256 bits of entropy, the effective classical security is 256 bits (not 512) and the Grover figure is 2^{128} iterations (not 2^{256}). The "key size" of MYSX is not the security level of the system. The 2^{512} and 2^{256} figures apply only if the caller supplies a uniformly random 64-byte key directly to `aeadEncrypt`.

7. Simon cryptanalysis: what transfers and what does not

MYSX uses Simon's `F`. It is **not** Simon. The differences:

- Block size and branch organization: MYSX is an SPN-like structure with four parallel Feistel cells over 32-bit word pairs (Section 2.1). Simon is a classical 2-branch Feistel over `n`-bit halves.
- Four parallel `F` per round (Simon uses 1).
- Myszkowski byte transposition (Simon has none).
- Round-dependent word rotations (Simon has none).
- Non-linear ARX key schedule (Simon's is linear).
- 128 rounds vs 68.

What transfers. The exact LAT of `F` — its closed form `|corr| = 2^{-h(b)}`, its validity condition `l ∈ colspace(A_b)`, and its sign structure — is a property of the function and therefore applies to MYSX regardless of the surrounding structure (Section 4.2 is

the transfer). That is the correct starting point for MYSX-specific linear cryptanalysis; the failed models of the v0.0 draft used a different (incorrect) F model.

What does not transfer. Key recovery from Simon's linear key schedule does not transfer: MYSX's key schedule is non-linear. Linear hull attacks against Simon are highly structured around Simon's Feistel-branch and its round-reduced behaviour; the MYSX hull is a different object and must be computed from the MYSX round model (Section 4.6). Integral distinguishers against Simon rely on Simon's particular 2-branch Feistel structure and its key schedule; the preliminary MYSX integral distinguishers in §4.9 are considerably shorter, and their relationship to Simon's is one of normalized coverage, not direct transfer.

Relevant literature on Simon (cited for context; the figures below were checked against the papers' abstracts, other figures quoted in earlier drafts were not verified and have been removed):

- Linear-hull key-recovery attacks with dynamic key-guessing reaching 49 rounds of Simon128/128 (Chen–Wang, "Improved Linear Hull Attack on Round-Reduced Simon with Dynamic Key-guessing Techniques", FSE 2016, ePrint 2015/666).
- Improved linear characteristics and hulls, including a 34-round linear characteristic with correlation 2^{-61} for Simon128 (Shi–Hu–Sun–Song– Qiao–Ma, "Improved Linear (hull) Cryptanalysis of Round-reduced Versions of SIMON", ePrint 2014/973).
- Differential analysis of Simon and Speck (Biryukov–Roy–Velichkov, "Differential Analysis of Block Ciphers SIMON and SPECK", FSE 2014).
- Integral distinguishers: 14 rounds for Simon32 with the conventional bit-based division property, and 15 rounds with three subsets (Todo–Morii, "Bit-Based Division Property and Application to Simon Family", FSE 2016); 14-, 16-, 18-, 22- and 26-round distinguishers for Simon32/48/64/96/128 (Xiang–Zhang–Bao–Lin, ASIACRYPT 2016).
- Structural observations on the SIMON family (Kölbl–Leander–Tiessen, "Observations on the SIMON block cipher family", CRYPTO 2015).

None of these results is transferred to MYSX in this paper. The MYSX-specific analysis (Sections 3, 4, 5) is what provides the evidence; Simon attacks are context, not direct transfer.

8. Analysis limitations

- **Differential**: upper bounds on single-trail for one input at r=12–14, no variation of the rotation offset; one trail verified at r=4 (ratio 2.00×); no lower bounds; no full clustering analysis; no inputs placed in the free windows.
- **Linear**: an exact model of `F` is established and matches the real cipher on small round counts, including exact signed hulls at r=10, 11, 12. A beam search over the exact model, **in both directions and merged over all 31 offsets**, up to 128 rounds (§4.10), gives achievable single-trail weights only: 2 at R=12, 5 at R=14, 38 at R=20, weight < 64 up to R=22 (63, a tie) and 2248 at R=128 (offset 3 backward; a real trail inside the cipher). The offset-20 backward trail at R=14 (rounds 100–113) is **isolated in its hull** (`hull_count` = 1) and its empirical corre-

lation matches 2^{-5}: that is a verified linear distinguisher on a 14-round window. Other trails cancel or are ≈2× larger, so single-trail weights are not universally distinguisher lengths. No lower bound over 128 rounds; no hull-aware bound beyond R≈14; `g` choices truncated to the 3 sparsest per cell; no key-recovery cost estimate. Multi-bit starts (pairs47, rand:5000:4) were explored and did not improve on single-bit starts, but the random sample is too small to be conclusive for structured multi-bit masks.

- **Integral**: preliminary. 6-round distinguisher with 2^{32} texts (word 0 active) and 5-round distinguisher with 2^{64} texts (words 0,4 active), both with the sound `(1,1)`-invalid division-property rule. A 10-round single-bit dependency propagation is reported for context but is not an integral distinguisher. Only r0 = 0 tested; other round windows, full-block inputs, and three-subset / traceability- enhanced DP are not explored. No key-recovery attack was attempted.
- **Related-key**: heuristic tests up to weight-6; no targeted MILP; number of round keys needed to recover the master key not measured.
- **Impossible differential, boomerang, algebraic**: not analyzed.
- **No independent public cryptanalysis.**
- **No post-quantum analysis** beyond the trivial Grover bound.
- **No side-channel analysis.**
- **Functional tests** (`mysx.test.js`, 17 tests) cover round-trip, AEAD negatives, nonce policy, PBKDF2, avalanche. They do not prove security.

9. Engineering notes

9.1. API surface

- Public API: `aeadEncrypt`, `aeadDecrypt`, `deriveAeadKeys`, `deriveAeadKeysFromPassword`, `NonceGuard`.
- Test API: **disabled by default**. Enabled only when the environment variable `MYSX_ENABLE_TEST=1` is set in the process before requiring the module. There is no code-level toggle.
- CTR (`_ctrXor`) is not exported from the public API.

9.2. NonceGuard

`NonceGuard` is a best-effort defense against nonce reuse. It:

- counts messages per key (default 2^{32});
- detects recent nonce reuse (window of 1024 per key);
- caps the number of tracked keys (default 1024, LRU eviction);
- identifies keys by a non-cryptographic 64-bit identifier made of two independent 32-bit lanes (a 32-bit FNV-1a lane and a multiplicative lane), **not** by the raw key bytes; `onSave(state)` therefore does not leak the secret key;
- throttles `onSave` calls (default: once every 100 `check()` calls) to avoid per-message state copies;
- **does not persist across restarts by itself**;
- **does not share state across processes or devices**.

Persistence is provided by `onSave(state)` and `initialState` at construction, or by `save()` / `load(state)` for manual control.

It is not a substitute for key lifecycle management. If the application reuses `(key, nonce)` across restarts, CTR confidentiality is broken. When a key is evicted from the tracked set its counter and nonce window are lost. See `mysx.prod.js` for the exact contract.

What it actually buys. `aeadEncrypt` draws a fresh 16-byte random nonce for every message, so an accidental repeat of `(key, nonce)` has probability about 2^{-65} even after 2^{32} messages under one key (birthday bound on 128 bits). The nonce-window check therefore guards mainly against a faulty random-number generator; the part that matters in practice is the per-key message limit, which is also lost when a key is evicted or the process restarts. The option name `unsafeNoGuard` reflects that the guard is missing, not that nonces are imposed by the caller (they are not: there is no public path to impose a nonce).

9.3. aeadEncrypt size validation

`aeadEncrypt` validates plaintext size **before** allocating the CTR output buffer, and enforces a configurable `MAX_PLAINTEXT_BYTES` limit (aligned with `MAX_CT_BYTES`, $2^{31}-1$). Applications that require larger limits should raise both limits explicitly and manage memory accordingly.

9.4. Type validation

All public functions validate that byte inputs are `Uint8Array` of the expected length (where applicable). Passing a `string` as `aad` or `plaintext` throws instead of silently coercing to array-like behavior.

9.5. KAT digests

Two digests are documented in `README.md`:

- `KAT.json` public digest: SHA-256 over the full JSON (block + AEAD + KDF). This is the artifact a third-party implementation should reproduce.
- `KNOWN_KAT_DIGEST` (internal): SHA-256 over 16 blocks with `key[i] = i`. Used as fast regression in `mysx.test.js`. It is strictly a subset of the public KAT, not a different KAT.

A third-party implementation should target the **public digest**.

9.6. Harness validation

The linear sampling harness is validated against a hand-computed case at w=8 (r=1, α = {word0 bit2, word4 bit0}, β = word0 bit3, fixed key). The linear approximation $x_0 \oplus F(x)_2 = x_1 x_6$ has correlation exactly ±1/2 with fixed key. Observed correlation: **0.4993** (expected 0.5, within sampling noise). With random keys per sample, the sign of the correlation flips per sample and the observed |corr| drops to **0.0022** (expected ≈ 0).

This confirms the harness is correct when used with fixed keys. It also confirms that the practice of "random keys per sample", used in an earlier version of `misc/mini_trail.js`, washes out genuine linear signals and must not be used for linear cryptanalysis.

See `misc/validate_lineal_harness.js`.

The mini-MYSX with `w=8, kw='MYSZKOWS'` is **degenerate**: the permutation becomes `[4,0,5,2,7,6,1,3]`, the L→R→L cycle at word 0 means a single-bit difference never passes through F, and the differential probability is 1 at all rounds. This is not representative of MYSX. Use `w=8, kw='MYSZ'` (non-degenerate from r=3) or w=32 directly. See `misc/check_mini_nondegenerate.js`.

The integral division-property harness (`misc/verify_division.py`) is validated against the R=10 single-bit dependency: the two bits the model predicts as balanced (word 0 bit 31, word 3 bit 6) are confirmed balanced over 20,000 random samples, and the other 62 bits of words 0 and 3 are confirmed not balanced. See `misc/verify_division.py` and §4.9.

The beam-search tool (`misc/mysx_linear_beam.py`) has its own `selftest` (LAT of F, one round against the real round function, 9-round weight-0 chains against the real cipher, forward/backward edge duality, backward 9-round weight-0 chains, backward R=14 trail is a valid forward trail) and a `reproduce` command that regenerates the §4.5 table. See §4.10.1.

10. Verdict

MYSX-256/512 v0.1-draft2 is an experimental proposal, not a secure cipher. The differential analysis reports upper bounds on single-trail weights (one input, rotation offset not varied), not a security proof; one trail at r=4 has been directly verified with clustering factor 2.00×. The linear analysis has a correct, exact model of `F` and of the round; it reproduces the real cipher's linear behaviour on the small round counts where it has been checked (verified `|corr| = 1` chains at 5 and 9 rounds; exact signed hulls at r = 10, 11, 12). A beam search over that model, run in **both directions** for all rotation offsets, up to 128 rounds, and merged, finds that the cheap region ends at roughly 20 rounds (single-trail weight 38 at R=20, below 64 only up to R=22 with weight 63 — a tie with the threshold — then growing by roughly 12–21 per round up to R=112; 2248 at R=128, offset 3 backward, which is a real trail inside the cipher). The offset-20 backward trail at R=14 (weight 5, rounds 100–113) is **isolated in its hull** (`hull_count` = 1) and its empirical correlation matches the prediction: **this is a verified linear distinguisher on a 14-round window**, made of five cheap rounds and a 9-round weight-0 chain. For other trails, a check against the real cipher shows that a single-trail weight can be exact, ≈2× too low, or exactly cancelled, so these numbers are evidence about where cheap trails end, not universally how long a distinguisher is. No lower bound over 128 rounds has been established, no key-recovery attack has been attempted, and hull-aware bounds beyond R≈14 are unexplored. Multi-bit starts (pairs47, rand:5000:4) did not improve on single-bit starts, but the random sample is too small to be conclusive. The integral analysis is preliminary: it establishes 5–6 round integral distinguishers with multi-bit inputs (2^{32} and 2^{64} texts) and a

10-round single-bit dependency propagation, but only for the first round window (r0 = 0). The related-key analysis is heuristic. No impossible- differential, boomerang, or algebraic analysis has been done. The cipher has not been externally audited.

Do not use MYSX to protect real data. For production: **AES-256-GCM** or **XChaCha20-Poly1305**.

11. Reproducibility

File	Purpose
`SPEC.md`	Formal specification (identical content to Section 2)
`PAPER.md`	This document
`KAT.json`	Test vectors (block, AEAD, KDF)
`generate_kat.py`	KAT generator
`mysx.prod.js`	Implementation (public API)
`mysx.test.js`	Functional tests
`cryptanalysis/models/linear/mysx_linear_model.py`	Exact linear model of F + round; `selftest` / `report` commands
`cryptanalysis/models/linear/LINEAR_MODEL.md`	Detailed documentation of the exact linear model
`cryptanalysis/models/linear/linear_model_data.json`	All data in Section 4 (except §4.10) as JSON
`misc/mysx_linear_beam.py`	Beam search over the exact linear model, all offsets, up to 128 rounds, forward and backward (§4.10); `selftest`, `reproduce`, `run`, `merge`
`misc/beam_full.log`, `misc/beam_full.json`	Output of the forward §4.10 run (`run --beam 2000 --jobs 8`)
`misc/beam_bwd.log`, `misc/beam_bwd.json`	Output of the backward §4.10 run (`run --direction backward --beam 2000 --jobs 8`)
`misc/beam_pairs47.json`	Output of the pairs47 multi-bit run (merged; not an improvement)
`misc/verify_beam_trail.py`	Prefix-by-prefix check of a saved beam trail against the real cipher, both directions (§4.10.3)
`misc/hull_count.py`	Counts trails sharing the end masks of a saved beam trail (§4.10.3)
`misc/mysx_milp.py`	Differential MILP
`misc/mysx_linear_milp.py`	Previous approximate linear MILPs (historical, superseded by `mysx_linear_model.py`)
`misc/mysx_division_property.py`	Bit-based division-property MILP (v1; `--xor-rule` selects XOR or OR)

File	Purpose
misc/mysx_division_property_v2.py	Bit-based division-property MILP (standard (1,1)-invalid XOR rule)
misc/verify_division.py	Empirical verification of integral distinguishers
misc/keyschedule_test.js	Basic key schedule audit
misc/keyschedule_test_extended.js	Exhaustive audit (22 tests)
misc/check_trail.js	Empirical differential trail at r=4
misc/check_linear_trail2.js	Empirical check of the superseded linear model (negative result)
misc/validate_lineal_harness.js	Hand-computed linear harness check
misc/mini.js	Mini-MYSX (w=8, w=16, w=32)
misc/check_mini.js	Mini-MYSX validation vs mysx.prod.js
misc/check_mini_nondegenerate.js	Keyword/w compatibility check
misc/mini_linear_fast.js	Linear sampling at w=8, kw='MYSZ'
misc/verify_trail_w32.js	Per-trail verification at w=32 with MILP weight
misc/avalanche.js	Diffusion and avalanche ladder (r=1..24)

Fixed seed in tests (0xC0FFEE). MILPs are deterministic given the solver and gapRel. The mini-MYSX with w=32 reproduces mysx.prod.js exactly (F function and MYZ_PERM are bit-identical, verified in misc/check_mini.js). The exact linear model's selftest additionally verifies the Python port against KNOWN_KAT_DIGEST of mysx.prod.js, so the linear results are stated relative to the reference implementation.

Beam search (§4.10). The forward run is reproduced with python3 misc/mysx_linear_beam.py run --beam 2000 --jobs 8 --out beam_full.json; the backward run with the same command plus --direction backward --out beam_bwd.json. The combined table is reproduced with python3 misc/mysx_linear_beam.py merge beam_full.json beam_pairs47.json beam_bwd.json. The search is deterministic given the parameters (no randomness except a fixed seed inside the coset sampler for cells with rank(A_b) > 8). cmp confirms that two independent forward runs (beam_full.json and beam_full_1.json) are byte-identical. The hull checks of §4.10.3 use the saved R=14 trails with fixed keys (random.Random(seed) over 64 bytes, seed 1 and 2) and are not part of the tool; their scripts are misc/verify_beam_trail.py and misc/hull_count.py.

Division-property results (§4.9). All three input patterns (single bit, word 0, words 0+4) were run with --r0 0. The (1,1)-invalid XOR rule and the OR-conservative rule were checked to agree on the tested patterns. Per-round balanced-bit counts are in the terminal logs of the runs; the runtimes were 0.4 s to 4.8 ks with SCIP. The R=10 single-bit dependency was verified against the real cipher (20,000 random samples per bit). The R=6 word-0 integral was verified against the real cipher at the last round (R=6, balanced bits confirmed; R=7 saturated). The R=5 words-0,4 integral was not verified empirically because its 2^{64}-text integral cannot be approximated by 20,000 random samples.

12. Final warning

MYSX has not been externally audited. This paper is the author's self-analysis, not a certification. No result here replaces an independent cryptographic review.

Do not use MYSX to protect real data.